



Level 0x10

Future Work v2026



Topics

- College / Investing in Education
- Internships
- Games and Hacking



Cost of Bachelor's Degree

- University of Florida
 - Undergrad: \$212 / credit. $120 * \$__ = ______$
 - Graduate: \$530 / credit. $30 * \$530 = \$15,900$
- University of Central Florida
 - Undergrad: \$212 / credit. $120 * \$__ = ______$ (was \$67 in late 1990s)
 - Graduate: \$369 / credit. $30 * \$369 = \$11,070$
- University of Miami
 - Undergrad: \$2,310 / credit. $120 * \$__ = \$______$
- Florida Tech
 - Undergrad: \$21,710 / semester. $8 * \$______ = \$______$
- Eastern Florida State College
 - AA (1000/2000 level): \$104 / credit. $60 * \$__ = \$______$
- Books: \$ / yr
- Living Expenses: \$ / year



Cost of Bachelor's Degree

- University of Florida
 - Undergrad: \$212 / credit. $120 * \$212 = \$25,440$
 - Graduate: \$530 / credit. $30 * \$530 = \$15,900$
- University of Central Florida
 - Undergrad: \$212 / credit. $120 * \$212 = \$25,440$
 - Graduate: \$369 / credit. $30 * \$369 = \$11,070$
- University of Miami
 - Undergrad: \$2,530 / credit. $120 * \$2,644 = \text{\textcolor{red}{\$277,200}} \text{\textcolor{red}{\$303,600}} \$317,280$
- Florida Tech
 - Undergrad: \$22,980 / semester. $8 * \$22,980 = \text{\textcolor{yellow}{\$173,680}} \text{\textcolor{yellow}{\$178,400}} \$183,840$
- Eastern Florida State College
 - AA (1000/2000 level): \$104 / credit. $60 * \$104 = \$6,240$
- Books: \$1,000 / yr
- Living Expenses: \$12,000 / year



Bright Futures Scholarship

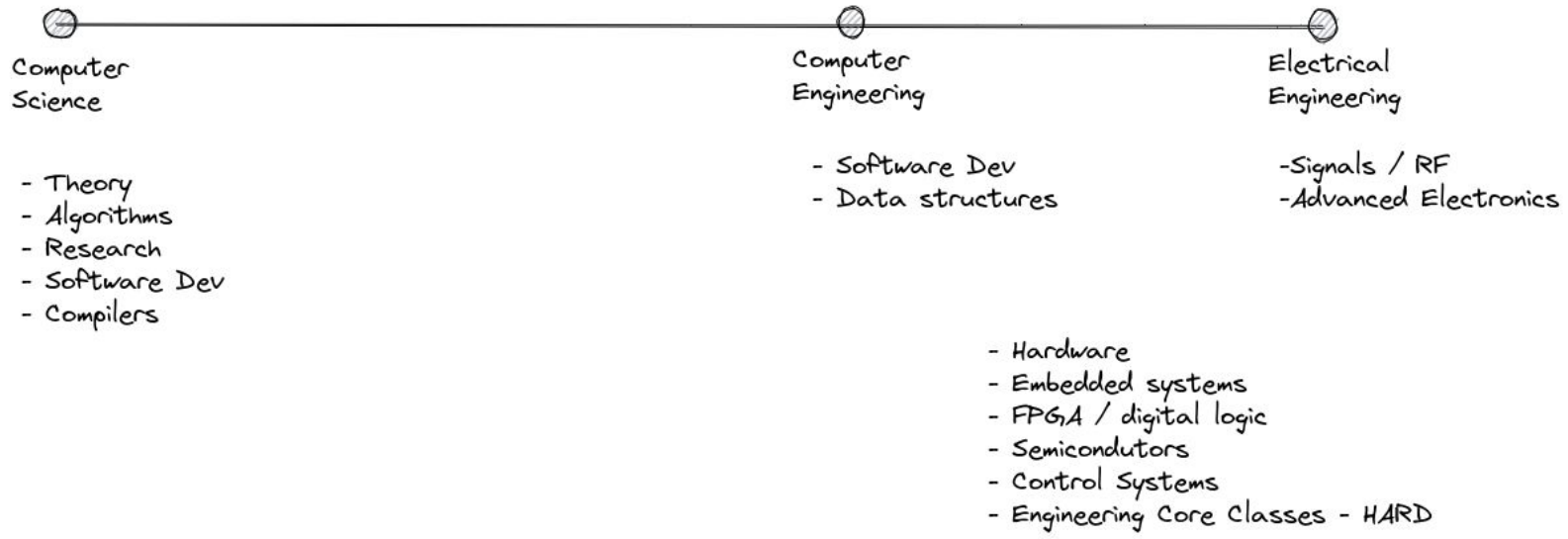
- Academic Scholarship
 - 3.50 GPA, 1330 SAT or 29 ACT
 - 100% of tuition at public university
 - \$6,350 / year
- Medallion Scholarship
 - 3.0 GPA, 1210 SAT or 25 ACT
 - 75% of tuition at public university
 - \$4,762 / year



Forbes 2026 Data

Job Title	Median / Top	Number of jobs	Est Growth
Product Marketing Mgr	\$160k / \$239k	434,000	+6%
Machine Learning Engr	\$141k / \$232k	40,300	+20%
Avionics System Mgr	\$135k / \$206k	71,600	+6%
Software Developer	\$131k / \$211k	1,895,500	+15%
IT Security Analyst	\$125k / \$186k	182,800	+29%
Electrical Engineer	\$119k / \$175k	287,900	+7%
Mechanical Engineer	\$102k / \$161k	293,100	+9%
Financial Advisor	\$102k / \$239k	326,000	+10%
Financial Analyst	\$101k / \$181k	429,000	+6%
Project Manager	\$101k / \$166k	1,046,300	+6%
UX Designer	\$95k / \$192k	214,900	+7%

Which Major? CS vs CpE vs EE



Nightwing CODEX Internships

NIGHTWING



- CODEX: Cyber Offense & Defense Experts
 - Multiple disciplines: software/hardware vulnerability research, reverse engineering, software engineering
 - What we do: computer network operations, computer network exploitation development, hardware emulation, system engineering, data analytics, and test engineering
 - We work on an array of technologies, including the latest mobile platforms, operating systems, embedded devices, and the newest data analytic platforms.
- [Nightwing Open Positions](#)
 - Summer internship positions will open/appear in January each year
 - Open to college students, any level
 - Offers 1 week PTO (paid time off)
- What to study?
 - Courses: Computer Architecture, OS, Networks, Crypto, Compilers
 - Languages: Python, C
 - CTFs
 - Debuggers and Binary Analysis (IDA, Ghidra, Binary Ninja)
 - Stack / heap based overflows



RIdiculously Awesome Cyber Internships

Our Cyber LOB supports Offensive Cyber Operations (OCO) missions. We solve challenging problems related to embedded-level software development, network protocols, RF interfaces, reverse engineering, vulnerability research, and computer network operations



- Why work here?
 - We create RIdiculously Awesome solutions that make the world safer.
 - Award Winning Culture - Seriously, awards were won
 - Interns receive 3 paid holidays
 - MentoRIng you won't receive anywhere else
- What we look for in an intern:
 - US Citizen
 - Typically pursuing a degree in Computer Science, Computer Engineering, or Cyber Security
 - Experience with vulnerability research and reverse engineering through online wargames, CTF competitions, or related platforms
 - Proficiency with assembly language, C/C++, Python, and fundamentals of operating systems
- Interns are on-site:
 - Melbourne, FL
 - San Antonio, TX
 - St. Petersburg, FL
- 3 interns each summer, for 3 months
- RII Careers Page:
 - <https://www.researchinnovations.com/careers>
 - Internships will be posted in the Fall



STR Internships

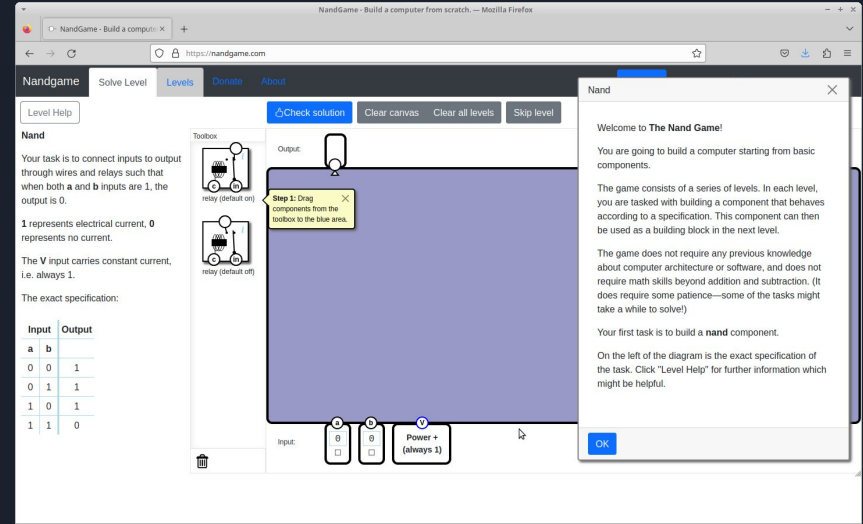


- What will the day-to-day work consist of during an STR internship?
 - Students will work on real programs with STR, no fake problems or concepts. We want our interns to gain real experience.
- When do the internship and co-op programs happen?
 - STR offers Spring and Fall Co-ops in addition to our Summer Internship program.
- <https://www.str.us/internships/> - Apply in January
- Who do we hire?
 - Electrical Engineering
 - Computer Engineering
 - Computer Science
 - Mathematics, Physics, Data Science and More

“Our full-time staff and interns practice: command and control, machine learning, cyber security, sensors development, signal processing, data analytics, & computer vision”

Digital Logic

- nandgame.com
 - Basic boolean algebra
 - Build primitive digital logic components
 - The building blocks of computers
- From Nand to Tetris
 - Book
 - Lecture Series (available on [Youtube](#))



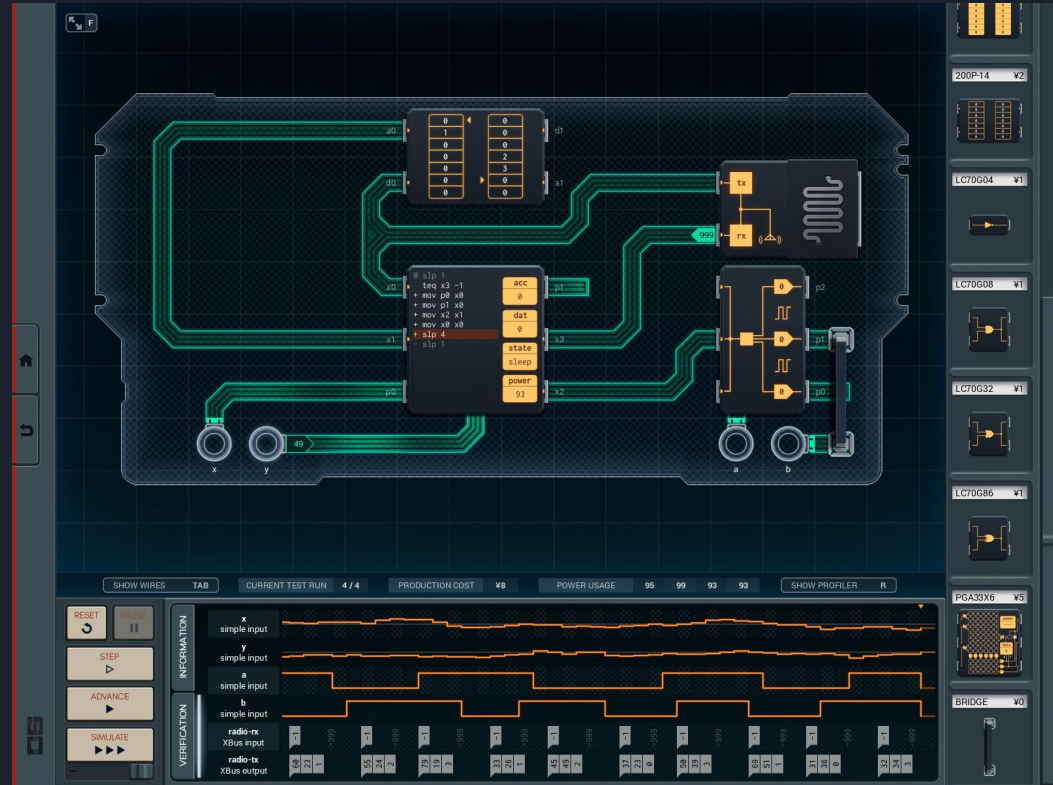
From Nand to Tetris

Noam Nisan and Shimon Schocken



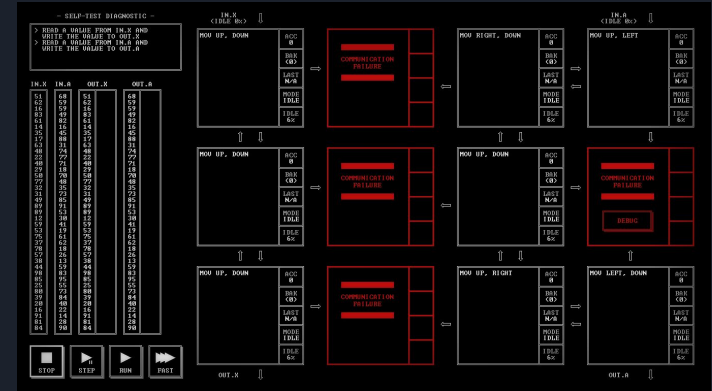
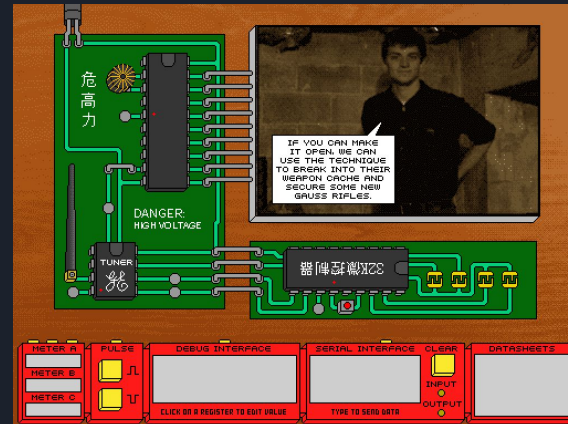
Shenzhen I/O

- PC / Steam Game
 - 95% positive rating
 - \$14.99
- Layout skills
- Assembly skills
- Timing diagrams
- Digital circuits



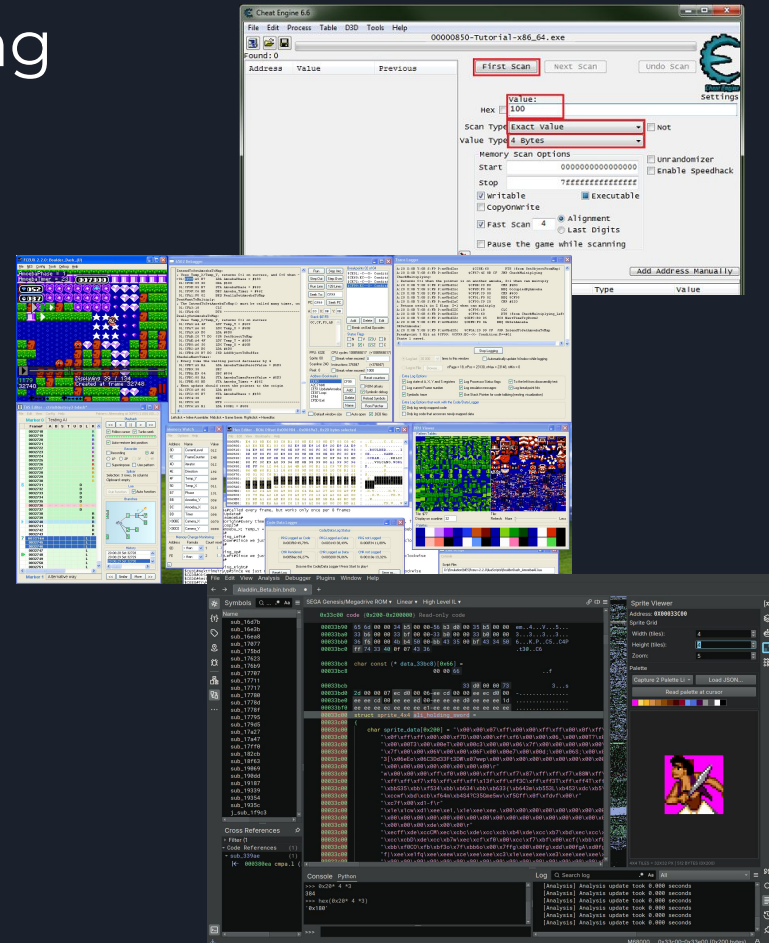
Other Zachtronics Titles

- TIS-100 (95% positive)
- Exapunks (96% positive)
- Last Call BBS (94% positive)
- Ruckingenur II
- KOHCTPYKTOP



Game Hacking / Modding

- Cheat Engine
 - Learn how memory can be manipulated
 - USE CAUTION
 - Some games may check for cheat engine
 - Don't get banned!
- Console Emulators
 - NES: fceux
 - Advanced debuggers

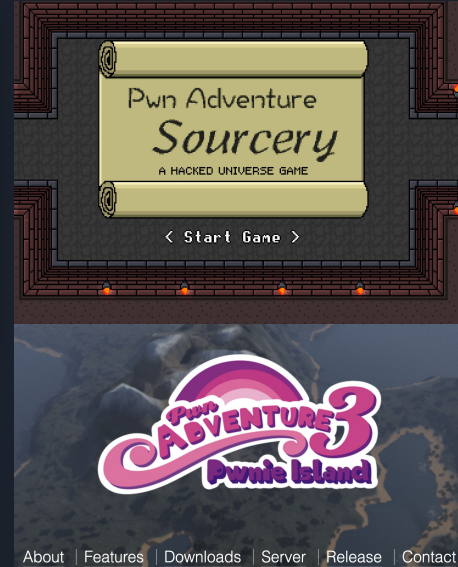


Hackable Games

PwnAdventure

- [PwnAdventure2](#)
- [PwnAdventure3](#)
- [PwnAdventureZ](#) (NES)
- [Online emulator](#)
- [SuperMonsterBall](#)
- [Sourcery](#) (Online)

[Squally](#)



CTFs

- pwn.college
- [picoCTF](https://picoCTF.org)
- Over the Wire
 - <https://overthewire.org/wargames/bandit/>
- HackTheBox.com
- [Cyber Quest Academy](https://cyberquestacademy.com) (Lockheed Martin)
- [Code Quest Academy](https://codequestacademy.com) (Lockheed Martin)
- [MicroCorruption](https://microcorruption.io) - MSP430

Disassembly

```
0010 < trap_interrupt>
0010: 3041      ret
4400 < _init_stack>
4400: 3140 0044  mov     #0x1400, sp
4404 < _low_level_init>
4404: 1542 5c01  mov     60x015c, r5
4408: 75f3      and.b   #0x1, r5
440a: 35d0 005a  bis     #0x5a08, r5
440e < _do_copy_data>
440e: 3f40 0000  clr     r15
4412: 0f93      tst      r15
4414: 0724      jz       $+0x10 < _do_clear_bss+0x8>
4416: 8245 5c01  mov     r5, 60x015c
441a: 2f83      decd    r15
441c: 9f4f 8645 0024  mov     0x4586(r15), 0x2400(r15)
4422: f923      jnz      $-0xc < _do_copy_data+0x8>
4424 < _do_clear_bss>
4424: 3f40 0000  clr     r15
4428: 0f93      tst      r15
442a: 0078      tv       Kahno, r5, r15, r0, r1, r2, r3, r4, r5, r6, r7, r8, r9, r10, r11, r12, r13, r14, r15, r16, r17, r18, r19, r20, r21, r22, r23, r24, r25, r26, r27, r28, r29, r30, r31, r32, r33, r34, r35, r36, r37, r38, r39, r40, r41, r42, r43, r44, r45, r46, r47, r48, r49, r50, r51, r52, r53, r54, r55, r56, r57, r58, r59, r60, r61, r62, r63, r64, r65, r66, r67, r68, r69, r70, r71, r72, r73, r74, r75, r76, r77, r78, r79, r80, r81, r82, r83, r84, r85, r86, r87, r88, r89, r90, r91, r92, r93, r94, r95, r96, r97, r98, r99, r100, r101, r102, r103, r104, r105, r106, r107, r108, r109, r110, r111, r112, r113, r114, r115, r116, r117, r118, r119, r120, r121, r122, r123, r124, r125, r126, r127, r128, r129, r130, r131, r132, r133, r134, r135, r136, r137, r138, r139, r140, r141, r142, r143, r144, r145, r146, r147, r148, r149, r150, r151, r152, r153, r154, r155, r156, r157, r158, r159, r160, r161, r162, r163, r164, r165, r166, r167, r168, r169, r170, r171, r172, r173, r174, r175, r176, r177, 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r2008, r2009, r2010, r2011, r2012, r2013, r2014, r2015, r2016, r2017, r2018, r2019, r2020, r2021, r2022, r2023, r2024, r2025, r2026, r2027, r2028, r2029, r2030, r2031, r2032, r2033, r2034, r2035, r2036, r2037, r2038, r2039, r2040, r2041, r2042, r2043, r2044, r2045, r2046, r2047, r2048, r2049, r2050, r2051, r2052, r2053, r2054, r2055, r2056, r2057, r2058, r2059, r2060, r2061, r2062, r2063, r2064, r2065, r2066, r2067, r2068, r2069, r2070, r2071, r2072, r2073, r2074, r2075, r2076, r2077, r2078, r2079, r2080, r2081, r2082, r2083, r2084, r2085, r2086, r2087, r2088, r2089, r2090, r2091, r2092, r2093, r2094, r2095, r2096, r2097, r2098, r2099, r2100, r2101, r2102, r2103, r2104, r2105, r2106, r2107, r2108, r2109, r2110, r2111, r2112, r2113, r2114, r2115, r2116, r2117, r2118, r2119, r2120, r2121, r2122
```


Coding Sites

- Codewars.com
- Codeingame.com
- codingbat.com
 - Java
 - Python

Or just write some code for a project you are interested in, put it on github!

Logic gates

All input signal data always have the same length.

The type of gates are:

- AND: performs a logical **AND** operation.
- OR: performs a logical **OR** operation.
- XOR: performs a logical **exclusive OR** operation.
- NAND: performs a logical **inverted AND** operation.
- NOR: performs a logical **inverted OR** operation.
- NXOR: performs a logical **inverted exclusive OR** operation.

Signals are represented with **underscore** and **minus** characters, an **underscore** matching a low level (0, or false) and a **minus** matching a high level (1, or true).

Input

Line 1: **n** number of input signals.
Line 2: **m** number of output signals.
next lines: two space separated strings: name of input signal, then signal form.
next lines: four space separated strings: name of output signal, then type of logic gate, then first input name, then second input name.

Output

m lines: two space separated strings: name of output signal, then signal form.

Constraints

$$1 \leq n \leq 4$$
$$1 \leq m \leq 16$$

Example

Input	Output
2	C
A	D
A	E
B	
C AND A B	
D OR A B	
E XOR A B	

```
C++
17 int m;
18 cin >> m; cin.ignore();
19 for (int i = 0; i < m; i++) {
20     string input_name;
21     string input_signal;
22     cin >> input_name >> input_signal; cin.ignore();
23 }
24 for (int i = 0; i < m; i++) {
25     string output_name;
26     string type;
27     string input_name_1;
28     string input_name_2;
29     cin >> output_name >> type >> input_name_1 >> input_name_2; cin.ignore();
30 }
31 for (int i = 0; i < m; i++) {
32     // Write an answer using cout. DON'T FORGET THE "<< endl"
33     // To debug: cerr << "Debug messages..." << endl;
34     cout << "output name and signal" << endl;
35 }
36 }
```

TEST CASES

TEST CASES	CUSTOM	ACTIONS
01	2 input signals, 3 gates	PLAY TESTCASE
02	Inverter (NAND)	PLAY TESTCASE

SUBMIT

Other Stuff

Vim-adventures.com

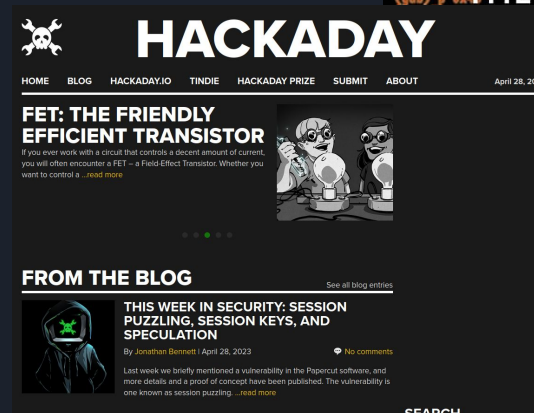
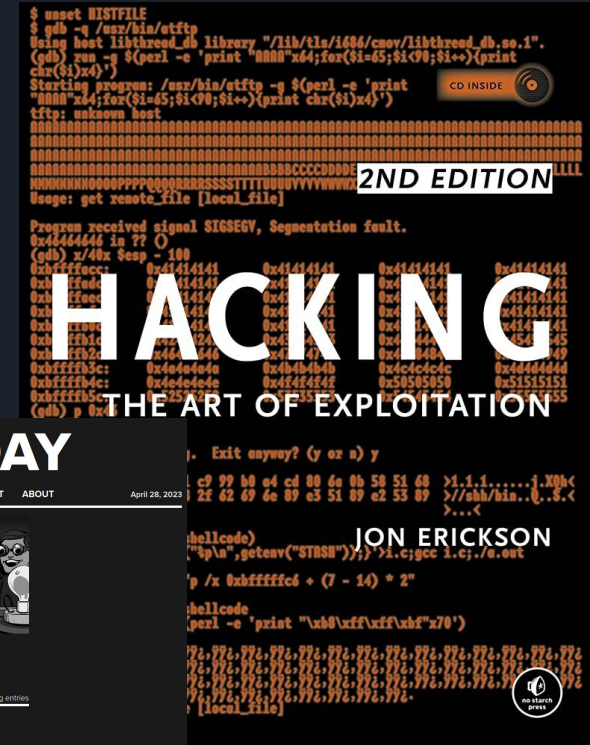
Why learn vim?

Do you T-Mux?



Reading

- Hacking: Art of Exploitation
 - C programming
 - Exploitation / bugs
 - If only there was a way to search for PDFs...
- Hack-a-day
 - Web blog
 - Retro projects
 - Embedded projects



Youtube

- [Matt Brown](#)
- [RE//verse Playlist](#)
- [DEFCON Presentations](#)
- [Chaos Computer Club \(CCC\)](#)
- [Live Overflow](#)
 - Hacking, exploitation, CTF
- Adafruit Industries
 - Limor Fried - Lady Ada
 - Making, electronics, microcontrollers
- John Hammond
 - CTF walkthroughs
- Mark Rober
 - Engineering, making
- Laurie Wired





Make something / have a project

- Something that wastes your time but you could improve with software
 - Repetitive tasks / boring work
- What new technology just became popular / affordable?
 - Wireless communications / mobile connectivity / mesh networks
 - AI / Machine learning
- Information processing / aggregation
 - Lots of websites have free APIs
 - Combine 2 different things together
- Make a game / rom hack / mod to existing game, or a demo
- Look for coding jams / hack-a-thons
- What could help a Ukrainian right now?

Keep Pwning!

pwn.college

Start Here

14 Hacking
2 Modules
10 Challenges



Linux Luminarium

59 Hacking
17 Modules
128 Challenges



Computing 101

77 Hacking
10 Modules
94 Challenges



Playing With Programs

14 Hacking
4 Modules
116 Challenges



Intro to Cybersecurity

53 Hacking
7 Modules
183 Challenges



Program Security

26 Hacking
5 Modules
159 Challenges



System Security

5 Hacking
6 Modules
93 Challenges



Software Exploitation

14 Hacking
5 Modules
90 Challenges



Westshore Jr/Sr High CS Club Dojo

Rank		Hacker	Badges		Score
#1		Impaler			47
#2		Somebody32x2			47
#3		Phoenix			11
#4		Guest102030			10
#5		evipu1l			8
#6		ZacharyW			6
#7		ianfburgospr			4
#8		cybergirl25			2



Links

- <https://fred.stlouisfed.org/>
- <https://www.nand2tetris.org/>
- <https://www.forbes.com/sites/bryanrobinson/2026/02/14/15-high-earning-bachelor-degree-jobs-most-paying-six-figures/>